Journal of Vibration Engineering & Technologies (2025) 13:590
https://doi.org/10.1007/s42417-025-02144-6

ORIGINAL PAPER

An Industrial IoT Framework for Predictive Maintenance of CNC Lathe Spindles: Integrating Deep Learning and Cloud-Based Analytics

Nikhil M. Thoppil[1] · **V. Vasu**[2]

Received: 5 May 2025 / Revised: 16 July 2025 / Accepted: 29 September 2025 / Published online: 11 November 2025
© Springer Nature Singapore Pte Ltd. 2025

Abstract

Purpose This study aims to develop an Industrial Internet of Things (IIoT)-based framework that integrates deep learning and cloud analytics for the predictive maintenance of CNC lathe spindle units. It addresses the challenge of vibration signature feature extraction and selection for the early detection of spindle degradation to optimize maintenance scheduling, improve product quality, and minimize downtime in industrial environments.

Methods An accelerated run-to-failure experimental testbed was developed to acquire spindle degradation data using vibration sensors. Neighborhood Component Analysis (NCA) was employed for feature selection. Long Short-Term Memory (LSTM) and bi-directional LSTM (bi-LSTM) deep learning models were trained using the processed time-series data for Remaining Useful Life (RUL) estimation. The framework was implemented on the MATLAB-powered ThingSpeak™ IoT platform, enabling real-time data acquisition, cloud analytics, and remote visualization.

Results The proposed system successfully predicted the spindle unit's RUL with high accuracy and robustness. Among the tested architectures, the LSTM model demonstrated the most balanced and consistent performance, achieving an RMSE of 40.01 time-steps and high A20-index values of 97.30% and 92.57% across the two test datasets. While the hybrid LSTM + bi-LSTM model achieved the lowest RMSE of 31.65 time-steps and an A20-index of 99.72% on one test set, its lower performance on the other set highlighted limitations in generalization. The cloud-based decision support system generated timely failure alerts, while the user dashboard provided continuous health status updates.

Conclusion The study demonstrates a viable IIoT-based predictive maintenance decision-making framework CNC lathe spindle unit. By integrating advanced sensor-driven data acquisition, deep learning algorithms, and cloud-based analytics, this framework supports proactive maintenance strategies through predictive analytics.

Keywords Predictive maintenance · Industrial Internet of Things (IIoT) · CNC lathe · Deep learning · Cloud analytics · Long Short-Term Memory (LSTM)

Introduction

Manufacturing systems are increasingly designed as highly sophisticated machine tools with advanced Computer Numerical Control (CNC) capabilities to remain competitive in the global market. The reliability of these manufacturing systems is directly influenced by the operating conditions of the machine tool's functional components and subsystems. As the industry continues to adopt more digital technologies, the absence of an effective maintenance strategy for machine tools could lead to significant challenges [1, 2]. Due to continuous operation and the nature of their work, machine tools experience wear and tear on their sliding and rotating components, which causes gradual mechanical damage. Such damage to critical machine tool components adversely impacts the quality of machined products and overall productivity [3, 4]. Therefore, an efficient machinery health management strategy is essential to minimize unexpected failures and reduce machinery downtime in the manufacturing industry [5].

The predictive maintenance approach enables users to effectively plan maintenance activities, ensuring the right actions are taken at the optimal time with minimal

✉ Nikhil M. Thoppil
nikhilmthoppil@mgits.ac.in

1 Muthoot Institute of Technology and Science, Ernakulam, Kerala, India

2 National Institute of Technology, Warangal, Telangana, India

production loss and expenditure [6, 7]. Murtaza et al. [8] examine the transition from Industry 4.0 to Industry 5.0 in predictive maintenance and condition monitoring, highlighting the challenges and opportunities in integrating advanced technologies for enhanced industrial performance. The predictive maintenance paradigm, a key component of Industry 4.0, is supported by several enabling technologies, including a wide array of sensors that capture various sources of information (such as vibrations, acoustic emissions, and temperature) from operating machinery. These sensors are complemented by Internet of Things (IoT) and Industrial Internet of Things (IIoT) - enabled remote connectivity, big data cloud storages, advanced computational resources for data analysis, and computing technologies. Together, these technologies provide real-time updates on machinery status, facilitating accurate prognostic analysis [9–11]. Nunes et al. [12] has reviewed systematically analyzes the challenges in predictive maintenance, emphasizing technical, organizational, and data-related aspects to guide effective implementation in manufacturing systems.

The Internet of Things (IoT), a cornerstone of Industry 4.0, enables the digital representation and interconnection of physical devices through unique identities, sensor systems, and communication protocols. It facilitates real-time data acquisition, integration, and analysis, transforming operational data into actionable insights across domains like manufacturing, healthcare, agriculture, and smart cities [13–16] In industrial contexts, this paradigm evolves into the Industrial Internet of Things (IIoT), which integrates smart machinery, cloud platforms, and AI-driven analytics to enhance equipment efficiency, productivity, and reliability. IIoT architectures typically include sensors, gateways, cloud storage, and intelligent user interfaces, working together to enable condition monitoring, fault detection, and data-driven decision-making. While the transformative potential of IIoT in predictive maintenance is widely recognized [17–19], challenges persist in system-level integration and deployment. To address these, recent studies have emphasized combining IIoT with AI and big data analytics to support advanced maintenance strategies that reduce downtime, optimize scheduling, and improve overall asset performance [20–23].

The rapid advancements in IoT and IIoT technologies have paved the way for deep learning to emerge as a powerful tool for analyzing time-series data in industrial applications. By enabling machinery health monitoring, failure prediction, and remaining useful life (RUL) estimation, deep learning is transforming how industries leverage data to enhance equipment reliability and performance. Through several layers of non-linear processing units, the deep learning models employ a hierarchical approach that allows them to learn complex patterns from both labeled and unlabeled data, making them well-suited for extracting valuable insights from diverse datasets. Widely used in fields such as computer vision, natural language processing, speech recognition, and time-series classification, deep learning has proven effective in extracting valuable insights from diverse datasets [24–27].

The Long Short-Term Memory (LSTM) network, an advanced form of recurrent neural networks (RNNs), excels in mining critical information from sequential time-series data [24, 28–31]. Generally, the deep learning network has a natural structure to learn machinery degradation patterns directly from raw machinery monitoring data. However, it demands a large size machine degradation data with multiple sensor observations to avoid the problem of underfitting during RUL estimations [32, 33]. Qi, et al. [34], Pie, et al. [35], and He et al. [36] have processed the machine degradation data to extract signature features representing the machine degradation trend, which is then utilized for training the LSTM deep network for accurate RUL estimation. The LSTM network's ability to learn directly from raw sensor data reduces computational complexity; however, its requirement for large volumes of machinery degradation data remains a significant limitation for industrial implementation [37, 38].

Despite considerable progress in the application of deep learning and IoT technologies for machinery health monitoring, most conventional models are either restricted to offline analysis or lack the integration needed for end-to-end predictive maintenance in real-time industrial settings. These approaches often emphasize isolated components such as sensor-level data processing or algorithmic modeling without offering a unified system that bridges data acquisition, intelligent prognostics, cloud-based analytics, and actionable decision-making. Additionally, many studies rely on simulated or benchmark datasets rather than experimentally validated degradation data, which limits their real-world applicability. The absence of real-time deployment platforms, modular architectures, and publicly accessible datasets further hinders reproducibility and practical adoption. This study addresses these limitations by developing and implementing a complete IIoT-enabled predictive maintenance framework that combines real-time vibration data acquisition, optimized feature selection, deep learning–based RUL estimation, and cloud-integrated decision support for CNC lathe spindle units. Lathe spindle degradation presents unique challenges due to its progressive and nonlinear nature, high-speed rotational dynamics, and sensitivity to surface finish and dimensional accuracy. The early symptoms are often subtle and difficult to detect, making timely fault identification and reliable RUL prediction complex yet crucial. A prior criticality analysis by Thoppil et al.

[39] identified the spindle unit as the most failure-prone and critical subsystem of a CNC lathe.

The remainder of this manuscript is organized as follows: Section Research Significance presents the Research Significance, highlighting the limitations of existing approaches and the contributions of the proposed framework. Section IIoT System Architecture details the architecture of the IIoT-enabled predictive maintenance system, while Sect. Testbed Design and Health Degradation Data Acquisition explains the experimental testbed setup and data acquisition process. Section Methodology outlines the methodology, including data preprocessing, feature selection, deep learning model development, and cloud-based integration. Section Results and Discussion discusses the results, model performance, and user interface for real-time maintenance decision-making. Finally, Section Conclusion concludes the study and outlines potential directions for future work.

Research Significance

The increasing integration of IIoT technologies and artificial intelligence has significantly advanced predictive maintenance strategies, particularly for rotating machinery. However, several limitations persist in current research and industrial practices. Many existing studies are limited to offline or retrospective RUL estimations, with minimal implementation in real-time environments. Furthermore, most research efforts focus primarily on sensor data conditioning and feature extraction to support failure analysis and forecasting. While these contributions are valuable, they often fall short of delivering a comprehensive, end-to-end predictive maintenance framework that seamlessly integrates real-time data acquisition, intelligent analytics, and actionable decision-making within a connected IIoT ecosystem.

To overcome these limitations, the present study introduces a robust and fully integrated IIoT-enabled predictive maintenance framework tailored for CNC lathe spindle units. The proposed architecture combines advanced deep learning models, namely, LSTM and bi-directional LSTM, with real-time cloud deployment over the MATLAB-powered ThingSpeak™ platform. This integration enables continuous condition monitoring, cloud-based predictive analytics, and automated remote maintenance alerts. The framework also leverages an accelerated run-to-failure testbed to observe the vibration signals form a healthy to faulty state of the CNC lathe spindles followed by vibration signature feature extraction and NCA based optimal feature selection, ensuring that the deep learning models are trained on the most informative inputs to deliver accurate RUL estimations in real-world conditions.

This approach advances the state of the art by demonstrating a practically deployable, real-time predictive maintenance system for CNC lathe spindles. It effectively bridges the gap between experimental modeling and industrial implementation by validating deep learning–based prognostic models within a live IIoT environment. The modular nature of the proposed architecture allows it to be adapted to various machine components and scaled across diverse industrial setups.

IIoT System Architecture

The proposed IIoT architecture aim to seamlessly integrate real-time condition monitoring data analysis, an LSTM and bi-LSTM-based deep learning predictive analytics model, a machinery health status visualization dashboard, and a live failure warning initiation system. The machinery health degradation data is acquired from the lathe spindle unit using sensors and a data acquisition system. This data is then analyzed to extract valuable insights for the prognostic analysis. Intelligent prognostic models, based on LSTM and bi-LSTM deep learning algorithm, are trained using the extracted machinery health degradation data to predict the RUL of the lathe spindle unit.

Once the most effective RUL estimation model is selected, it is uploaded to the cloud storage for real-time predictive analytics. The real-time machinery health degradation data, representing the current operational state of the lathe spindle, is continuously acquired and made available in the Think Speak cloud analytics platform. An intelligent cloud-analytics algorithm employing the trained LSTM and bi-LSTM deep learning model is developed to estimate the RUL of the lathe spindle unit in real time. This information, hosted on the cloud analytics platform, is used to assess potential failures and trigger email alert warnings accordingly. Furthermore, the real-time machinery health degradation data is also displayed on a webpage visualization interface, providing a user-friendly view of the lathe spindle's health status. The real-time machinery health status information and failure warnings based on the estimated RUL can be put into use for making informed predictive maintenance decisions for the lathe machine tool. Figure 1 presents the outline of the proposed IIoT cloud analytics framework.

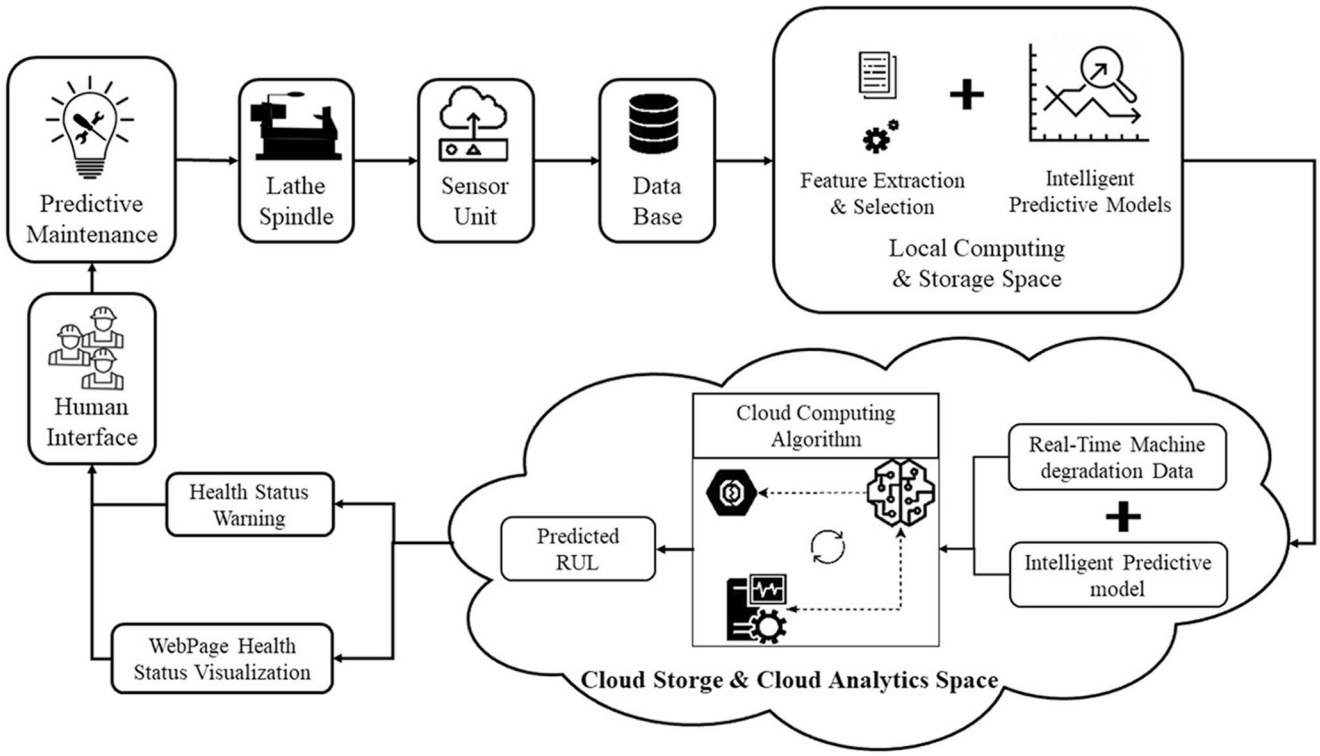

Fig. 1 Outline of IIoT cloud analytics framework

Testbed Design and Health Degradation Data Acquisition

An accelerated run-to-failure experimental test setup is fabricated to acquire spindle health degradation data. The experimental testbed primarily comprises a lathe spindle assembly, a drive motor with the belt-pulley arrangement, and a loading arrangement [40]. The speed (rpm) of the spindle unit is varied for different runs by varying the driver pulley ratio. The loading arrangement is placed replacing the chuck of the lathe spindle. The arrangement applies a constant radial load on the spindle and it also ensures that there is no shock load acting on the spindle. As in any mechanical system, the vibration signal is identified as the most suitable parameter for condition monitoring of the spindle unit testbed. 'PCB-603C01' accelerometer sensors are employed for vibration monitoring of the spindle unit [41]. The accelerometer model has a sensitivity of 100 mV/g, a frequency range of 0.5 to 10,000 Hz, and a measurement range of ± 50 g (g = 9.8 m/s2). An increase in overall vibration levels indicates the development of a fault or deterioration in the condition of machines. 'NI-9234' sound and vibration module with 'cRIO 9171' data acquisition (DAQ) system are used to record monitored vibration signals. The accelerometers are mounted directly on the top of spindle bearings and are connected to the data acquisition device. The data acquisition system acquires vibration data from the accelerometers and transfers it to the connected computer storage.

The entire experimental setup is assembled on a heavy machine worktable using anti-vibration gaskets to minimize the presence of noise in the condition monitoring data. The accelerated run-to-failure experimental testbed with component labels is shown in Fig. 2. The accelerometer data is acquired at a sampling rate of 25.6 kHz following a time-step of once every 60 s. Each of these acquired data packets represents the health state of the spindle unit at that time step. The machine health degradation data is represented by these data packets in their respective time-steps. Each degradation data represents the health degradation of the spindle unit from a healthy state to a faulty state.

The faulty state of any machine component is when it is not able to perform its operation in a predefined desires manner. The faulty state of a lathe spindle unit can be defined as the point at which it is unable to produce machined parts within the tolerance limits of surface finish and dimensional accuracy. Accelerometer reading (vibration amplitude) above 40 g is defined as the failure state of the experimental testbed considered, where g is the acceleration due to gravity 9.81 m/s2. The rotating element bearings in the lathe spindle assembly are very critical to health degradation failure [39].

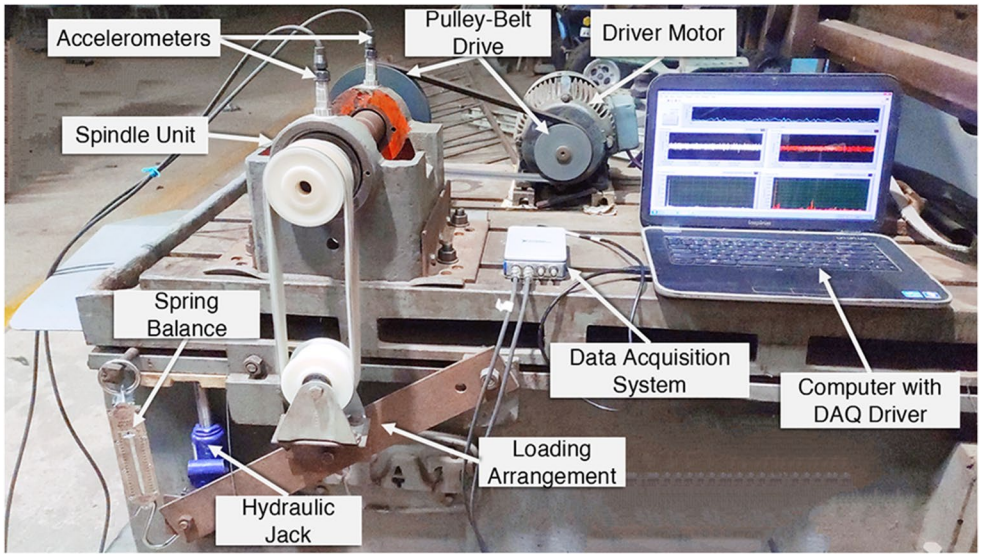

Fig. 2 Accelerated run-to-failure testbed design

These spindle bearings (32211, 32212 tapered roller bearings) are replaced with new bearings and the spindle unit is completely reconditioned before each failure runs.

The lathe spindle health degradation dataset includes 14 accelerated run-to-failure experiments conducted at a constant load and five carefully chosen spindle speeds: 823 rpm, 900 rpm, 1400 rpm, 1800 rpm, and 2520 rpm. These speeds were selected based on preliminary trials that evaluated the effectiveness of different operating conditions in generating observable degradation patterns. Rotational speeds below 823 rpm produced minimal variation in vibration signals even after extended operation, while speeds above 2520 rpm led to rapid and unstable signal fluctuations over a short duration. The five selected speeds thus represent a practical and experimentally stable range that captures the maximum useful variation in vibration signatures, enabling accurate modeling of spindle degradation behavior. Table 1 summarizes the corresponding spindle speed and the maximum operating life of each bearing unit in time-steps (one time-step equals 60 s).

Methodology

The proposed methodology is structured into three primary stages. First, a data-driven health degradation modeling and RUL estimation process is conducted. In this stage the collected data undergoes analysis and preprocessing to build an RUL prediction model. In the second stage, a cloud-based data management and predictive analytics framework is developed using the ThingSpeak™ IoT platform to enable real-time data processing and condition monitoring. Finally, a cloud computing architecture is established, incorporating a user-friendly interface that provides real-time health status updates and failure warning alerts.

Data-Driven Health Degradation Modeling and RUL Estimation

The data acquired from the experimental setup is analyzed to extract useful information on machinery failure. Further, data-driven techniques are employed to extract machine health degradation patterns to evolve intelligent predictive models.

The machinery health degradation data comprising at most a single vibration sensor data might cause underfitting of the deep learning model. Vibration signal signature features revealing superior machinery degradation patterns are considered for deep network training. The raw vibration signals are first subjected to wavelet denoising with 'db5' mother wavelet at level-4 to filter out high-frequency noise signals [42, 43]. Then, statistical features are extracted from the denoised signal using time, frequency, and time-frequency domain analysis. The features extracted from the

Table 1 Lathe spindle health degradation data with maximum life

S. No.	Speed (rpm)	Maximum Life (Time-Steps)
1	823	1942
2		1926
3		1816
4	900	1802
5		1674
6		1230
7	1440	1146
8		1072
9		972
10	1800	923
11		901
12		857
13	2520	647
14		624

denoised vibration signal include; mean, standard deviation, root mean square (RMS), root sum of square (RSSq), peak-to-peak, crest factor, impulse factor, margin factor, skewness, kurtosis, shape factor, mean frequency, time domain energy, wavelet energy, spectral entropy, spectral kurtosis, Shannon entropy, Log entropy, normal entropy, approximate entropy, joint moment, and mean peak frequency [44, 45]. Table 2 summarizes the statistical features with their computation formulae.

These extracted vibration features are concatenated to form a compound dataset, which is then subjected to an NCA-based feature selection criterion, where only those features are selected which could maximize the prediction accuracy of evolved predictive models. A regularized NCA

Table 2 Statistical features in time, frequency, and time-frequency domain

FI	Feature	Formula
1	***Mean***	$M = \frac{1}{N}\sum_{i=1}^{N} x_i$
2	***Standard deviation***	$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})}$
3	***RMS***	$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
4	***Impulse factor***	$IF = \frac{max\lvert x_i\rvert}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert}}$
5	***Skewness***	$Sk = \frac{\sum_{i=1}^{N}(x_i - m)^3}{(N-1)\sigma^3}$
6	***Shape factor***	$SF = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}}{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert}$
7	***Energy (Time domain)***	$E = \sum_{i=1}^{N} x_i^2$
8	***Spectral entropy***	$e(p) = -\sum_{i=1}^{n} p(x_i)\, log_2(p(x_i))$ *P(.)=probability function*
9	***Shannon entropy***	$sEntropy = -\sum_{i=1}^{N} x_i^2 \log x_i^2$
10	***Normal entropy***	$nEntropy = \sum_{i=1}^{N}\lvert x_i\rvert^p$ *p=signal threshold*
11	***Joint moment***	$momentJ = \iint t^n w^m P(t,\omega)\, dt d\omega$ m=*order*, *P(.)=marginal distribution*, *t=first temporal moment*, *ω=spectral time-freq. moment*
12	***Peak-to-peak***	$P2P = \max\lvert x_i\rvert - \min\lvert x_i\rvert$
13	***RSSq***	$RSSq = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert^2}$
14	***Crest factor***	$CF = \frac{max\lvert x_i\rvert}{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
15	***Margin factor***	$MF = \frac{max\lvert x_i\rvert}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert^2}}$
16	***Kurtosis***	$Ku = \frac{\sum_{i=1}^{N}(x_i - m)^4}{(N-1)\sigma^4}$
17	***Mean frequency***	$meanfreq = \frac{1}{N}\sum_{i=1}^{N} f_i$
18	***Wavelet energy***	$WE = \sum_{i=1}^{N}\frac{\omega\, t_{\varnothing}^2(i)}{N}$
19	***Spectral kurtosis***	$K(f) = \frac{\lvert S(t,f)\rvert^4}{(\lvert S(t,f)\rvert^2)^2} - 2,\ f \neq 0$ *S(.)=STFT of x_i, t=time, f=frequency*
20	***Log entropy***	$lEntropy = \sum_{i=1}^{N}\log x_i^2$
21	***Approx. entropy***	$AE = \varnothing_m - \varnothing_{m+1}$ $\varnothing_m = (N-m+1)^{-1}\sum_{i=1}^{N-m+1}\log(N_i)$
22	***Mean peak frequency***	$meanPeakFreq = \frac{1}{T}\int_0^T argmax_w(Sp(t,\omega))\, dt$ *Sp(.)=Spectrogram, ω=individual frequency*

***FI=Feature Index**

algorithm is executed using the compound feature dataset representing all 14 failure runs to determine the weighting vector corresponding to each feature. A relative threshold value for feature weights equals 15.0 is assigned from random trials as the cut-off criterion to select the most relevant feature subsets for all 14 failure datasets. All the remaining features are discarded at this stage of analysis.

RUL Estimation Using LSTM/bi-LSTM Deep Network Architectures

The computational algorithms for predictive modeling are the most crucial and challenging step in machine failure prognostics. The lathe spindle health degradation data is fed to train an LSTM, bi-LSTM, and hybrid LSTM + bi-LSTM deep network architectures to evolve intelligent predictive analytics models for the RUL estimation of the lathe spindle unit. Table 3 presents the layer framework of these deep learning architectures. A sequence input layer inputs sequence data to a network. The fully connected layer establishes the connection between every neuron in one layer of the LSTM/bi-LSTM network model to every neuron on preceding layers. The dropout layers regulate the learning process to avoid overfitting of LSTM/bi-LSTM network models [45]. A regression layer computes the error loss for regression tasks. The hybrid LSTM + bi-LSTM network architecture follows a similar layer framework where, the LSTM layer + fully connected layer is followed by a bi-LSTM layer + fully connected layer. Flow diagram for the LSTM and bi-LSTM network algorithm shown in Fig. 3.

The training and validation datasets are fed into the LSTM and bi-LSTM network algorithm to evolve accurate predictive models for RUL estimation. The dataset consists of 14 run-to-failure experiments, each capturing the complete degradation cycle of a CNC lathe spindle unit. For model development, 10 failure runs were randomly selected and used as the training dataset. The remaining four runs were evenly divided, with two runs allocated for validation and two for testing. This approach ensures that all subsets include full degradation trajectories, preserving the temporal progression patterns critical for accurate Remaining Useful Life (RUL) modeling.

The hyperparameters control the behavior of deep learning architecture. The most relevant hyperparameters involved in the training of LSTM/bi-LSTM deep learning architecture are number of hidden units, number of fully connected layer, dropout rate, maximum epochs, initial learning rate, and regularization factor [46]. The values for each hyperparameters are determined employing a Bayesian search optimization technique. Bayesian optimization of LSTM, bi-LSTM, hybrid LSTM + bi-LSTM network architectures are performed separately against the training and validation datasets to identify their best hyperparameters. Thoppil et al. [45] presented a Bayesian optimization framework for automated hyperparameter tuning and structural configuration of LSTM/bi-LSTM models. The algorithm trains the deep network models and returns the validation error after each iteration. The validation error is defined as the mean absolute error (MAE) between the predicted

Table 3 Layer framework of LSTM and bi-LSTM network architectures

Layer No.	LSTM	bi-LSTM
1	*Sequence input layer*	*Sequence input layer*
2	*LSTM layer*	*bi-LSTM layer*
3	*Fully connected layer*	*Fully connected layer*
4	*Dropout layer*	*Dropout layer*
5	*Fully connected output layer*	*Fully connected output layer*
6	*Regression layer*	*Regression layer*

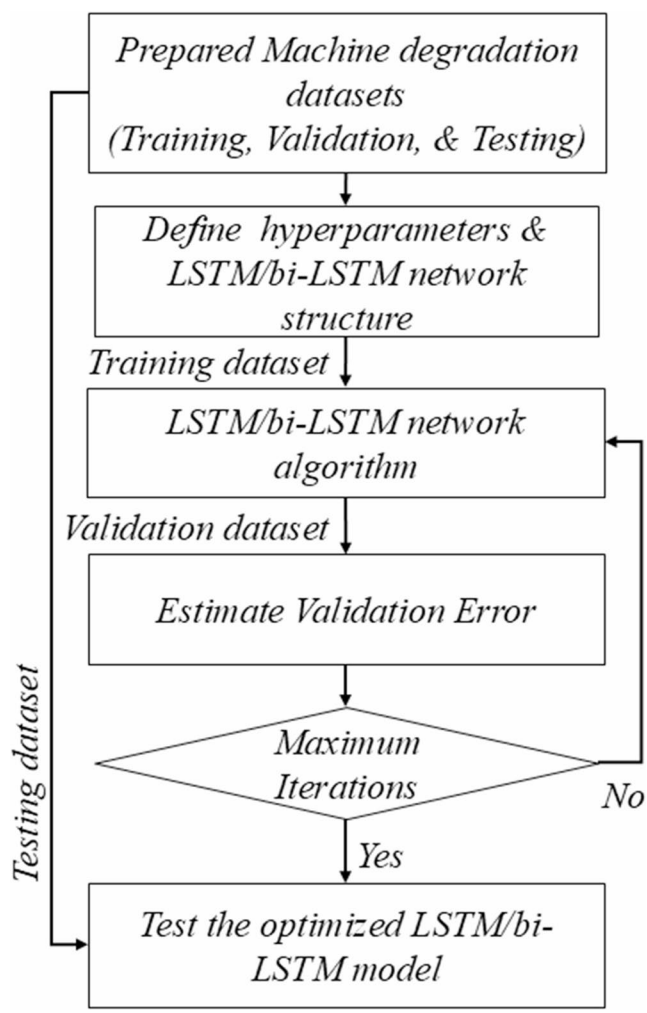

Fig. 3 Flow diagram for LSTM and bi-LSTM network algorithm

machine health degradation trend for the validation dataset and the actual machine health degradation trend as shown in Eq. (1) [45, 47].

$$Validation\ Error = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n} \tag{1}$$

where y_i is the predicted machine health degradation trend, x_i is the actual machine degradation trend, and n is the total time-steps of the validation dataset. The algorithm executes over iterations until the termination criteria is reached to minimize the validation error thus refining the predictive model.

The termination criterion for the algorithm is set to a predefined condition of completing 30 iterations. The condition is arbitrarily chosen based on random trials considering a parity between the computational expense and prediction accuracy. Finally, the evolved predictive models are further investigated to determine the prediction accuracy on an independent testing dataset. The root mean square error (RMSE) represents the difference between the actual and predicted RUL is estimated for the predictive models. RMSE is computed as shown in Eq. (2) [45, 47].

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - x_i)^2}{n}} \tag{2}$$

where n is total time-steps, y_i is predicted values, and x_i is actual values.

In the context of this research, which focuses on RUL prediction for CNC lathe spindle degradation, the A20-index serves as a complementary metric to RMSE. While RMSE measures average error magnitude, the A20-index captures how consistently the model maintains predictions within an acceptable range of the true values. This is especially important for condition monitoring and prognostics, where even a small number of large errors could have critical consequences. The application of the A20-index in this work not only strengthens the evaluation framework but also aligns the performance assessment with engineering relevance, making it easier to compare model robustness across varying test conditions and network architectures. A20-index defines the percentage of data points where the predicted values deviate by less than ±20% from the corresponding experimental values as shown in Eq. (3).

$$\text{A20} - \text{index} = \frac{m20}{M} \tag{3}$$

where, M is the total sample count for the dataset, and m20 is the total sample count for samples having an experimental value/predicted value ratio between 0.80 and 1.20 [48, 49].

Cloud-Based Data Management and Predictive Analytics Framework

The cloud-based data management and predictive analytics framework uses an IoT platform to communicate with the lathe spindle unit, monitor live data from local computing and storage systems, and manage historical data and evolved predictive models for real-time prognostic analysis. In this work, the ThingSpeak™ IoT platform by MathWorks® is employed for cloud computing integration. The local computing and storage system is connected to ThingSpeak™, where spindle real time health status is visualized and failure alerts are generated.

ThingSpeak is a cloud-based IoT analytics platform that enables users to aggregate, visualize, and analyze real-time data streams [50, 51]. ThingSpeak uses channels to store data transmitted from local storage systems or ground devices over the internet. Data can be sent to or retrieved from ThingSpeak channels using a REST API or MQTT API (REST- Representational state transfer, MQTT- Message Queue Telemetry Transport, API-Application Programming Interface). The REST API calls are used to create and update the ThingSpeak channels and the MQTT API is used to update the ThingSpeak channels. MQTT is an Organization for the Advancement of Structured Information (ISO) and International Organization for Standardization (OASIS) standards messaging protocol for the IoT. In addition, ThingSpeak also encourages cloud-to-cloud interactions, which is utilized in this work to obtain the evolved predictive models for RUL estimation.

The core element for communication in the ThingSpeak IoT web service is a ThingSpeak channel that establishes connectivity among the devices/things over the internet. ThingSpeak allows the user to store and retrieve data to and from the channel in real-time. The ThingSpeak channel has fields for data, fields for device station location, and fields for status for varied sensed data. REST API calls 'GET' and 'POST' can be used to send and retrieve data to and from the channel respectively. The MQTT 'Publish' method can also be used to update the channel fields and MQTT 'Subscribe' can be used to retrieve channel data. MATLAB functional codes are also employed to write or read data to and from the ThingSpeak channel. The ThingSpeak channel uses separate API keys namely, the 'Write API key' and 'Read API key' to write and read data from channel respectively. The data are stored in the ThingSpeak channel in either JSON, XML, or CSV data formats.

In the present work a MATLAB function code using the channel 'Write API key' is defined to upload the real-time lathe spindle health degradation data from the local storage space to the IoT ThingSpeak channel. The chosen vibration signature features are sent to the 'Lathe Spindle Health Status' monitoring channel employing the 'Write API key' function. The live data acquired from the lathe spindle test setup is first analyzed to extract the meaningful vibration signature feature and then selected features uploaded to the ThingSpeak data collection channel at a rate of once in every 60 seconds. The different vibration signature features are uploaded to individual fields of data defined in the ThingSpeak channel namely, standard deviation, peak-to-peak, RMS, RSSq, energy, normal entropy, and log entropy. Similarly, a MATLAB function code using the channel 'Read API key' is defined to retrieve the data from the channel when required for analysis. The data read from the channel fed in a cloud computing space where the live lathe spindle health degradation data is analyzed against the evolved predictive model for the RUL estimation of lathe spindle unit and further failure alert warning through email notifications. The evolved predictive model required for the RUL estimation is made available in the cloud space through the 'Drop Box' cloud storage service provider. The evolved intelligent LSTM predictive model is uploaded to the DropBox server and is called in the ThingSpeak cloud analytics space when required. Figure 4 shows the flow of data in the proposed IoT system.

ThingSpeak™ IoT web service offers a MATLAB-based cloud computing environment for analyzing channel data. The architecture consists of three main components: cloud storage, cloud computing, and a user interface. Figure 5 illustrates the flow diagram of the IoT cloud computing architecture and user interface developed for predictive analytics of the lathe spindle unit. The architecture and interface are discussed in detail in the following sections.

Results and Discussion

Feature Selection and Data Preparation

Figure 6 illustrates the NCA results, showing the weighting vectors and corresponding feature indices (FI), with the features listed in Table 2. A relative threshold value of 15.0, determined through random trials, is used as the cut-off criterion to select the most relevant features across 14 failure datasets. Features such as standard deviation, RMS, energy, normal entropy, peak-to-peak, RSSq, and log entropy exceed this threshold and are selected for RUL estimation using data-driven prognostic algorithms. The remaining features are discarded. Training the predictive models using only these high-weight features improves their prognostic accuracy.

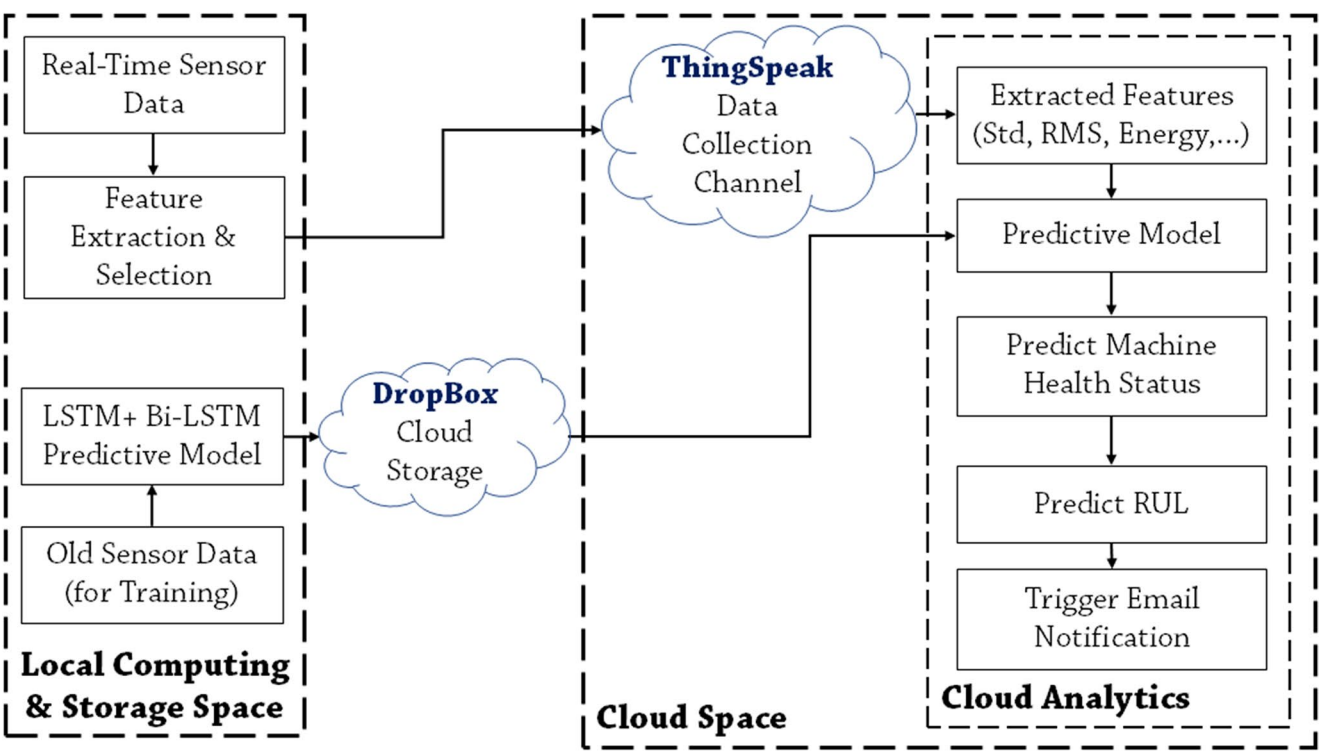

Fig. 4 Flow of data in IoT Platform

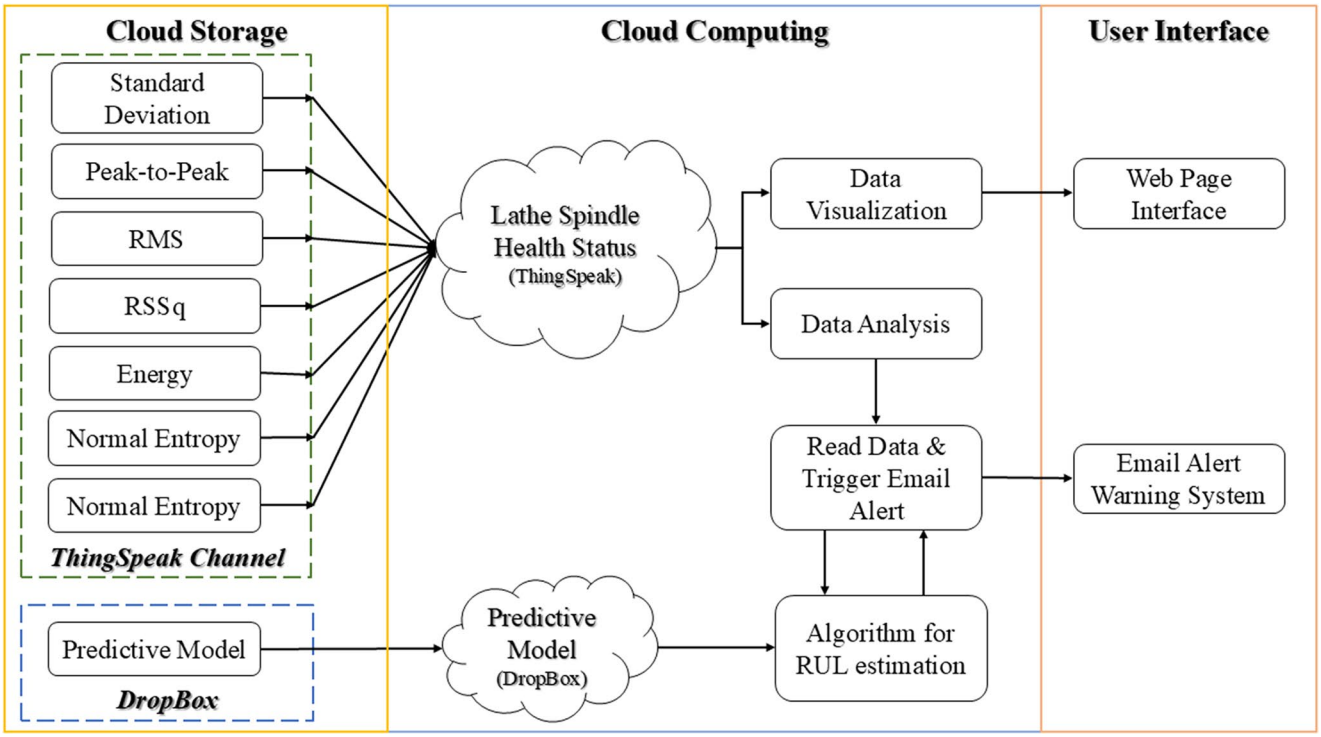

Fig. 5 IoT cloud computing architecture and user interface for the predictive analytics of lathe spindle unit

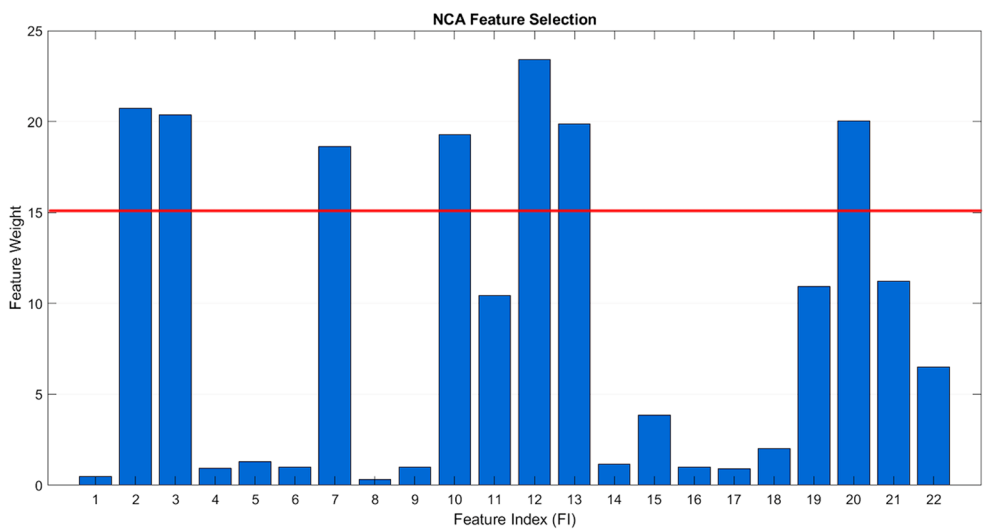

Fig. 6 NCA feature selection - Feature weights and corresponding feature index

For predictive analytics, the available dataset is divided into training, validation, and testing sets. The time-series training dataset is normalized to have zero mean and unit variance. The validation and testing datasets are normalized using the same mean and standard deviation as the training set.

RUL Estimation of Lathe Spindle Unit

The algorithms employed in this work are developed and executed on a computer equipped with an Intel Core-i7 (3.60 GHz) processor, 8 GB RAM, and running 64-bit MATLAB software. The LSTM, bi-LSTM, and hybrid LSTM + bi-LSTM deep network architectures are separately trained on lathe spindle health degradation data. The Bayesian optimization algorithm tune the hyperparameters towards the attainment of a minimum validation error. The prediction accuracy of the evolved predictive models is then compared to determine the most accurate and consistent predictive model.

Figures 7 and 8 provide a comprehensive visual analysis of the prediction performance for the three deep learning

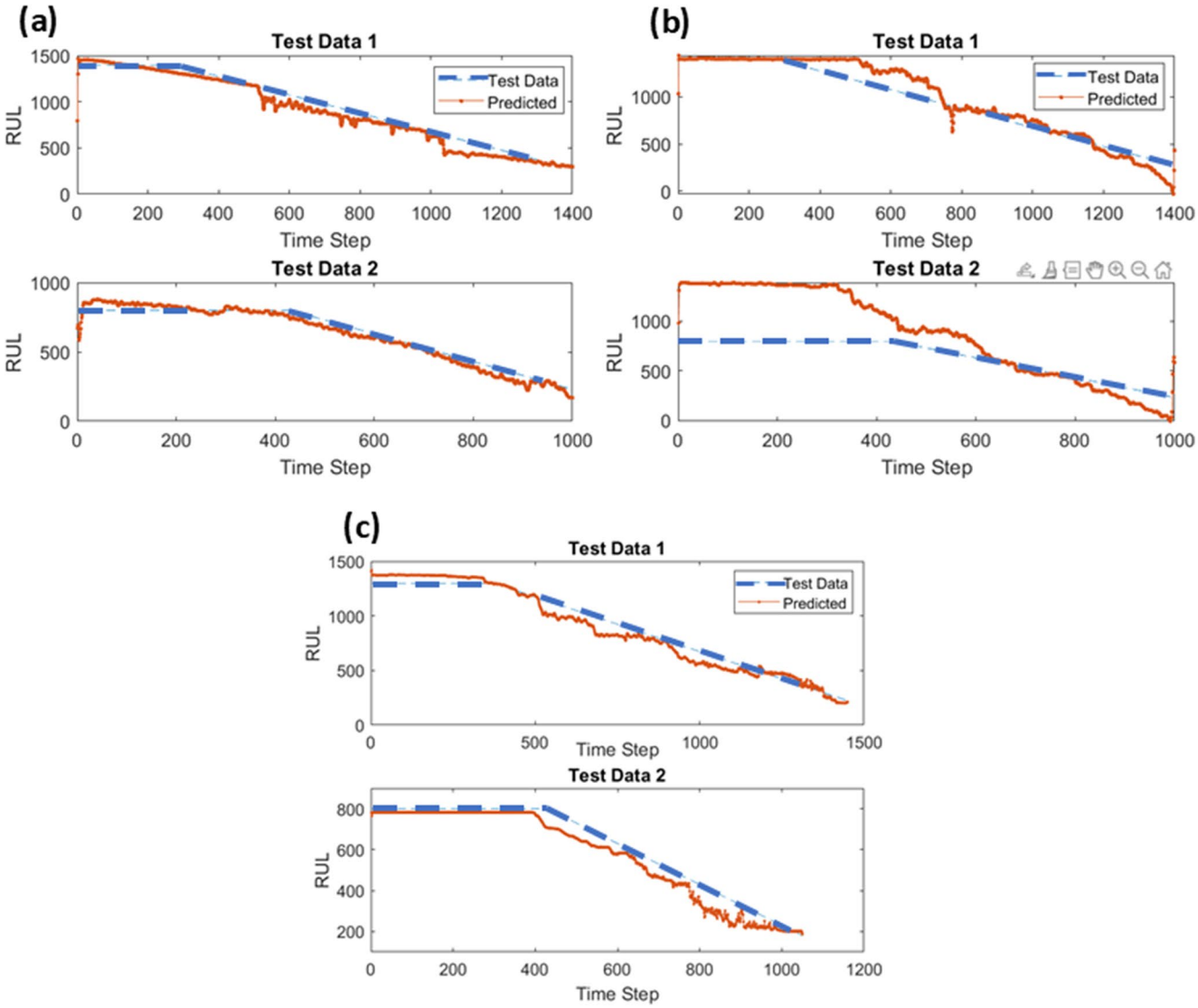

Fig. 7 Time-series comparison of actual and predicted RUL values (**a**) LSTM (**b**) bi-LSTM, and (**c**) LSTM + bi-LSTM network architecture

architectures LSTM, bi-LSTM, and hybrid LSTM + bi-LSTM on RUL estimation. Figure 7 illustrates the model predictions in terms of RUL versus time-step, where the actual RUL values from the test datasets are plotted alongside the predicted RUL values for each network. This representation allows for a time-series comparison, highlighting how closely each model tracks the degradation trend over time. Notably, the plots help identify prediction lag, early failure warnings, or overestimation tendencies.

In contrast, Fig. 8 presents predicted RUL values plotted directly against the corresponding true RUL values for both test datasets. Ideally, accurate predictions should lie along the 45-degree diagonal line, which represents a unit slope (slope = 1) and perfect agreement between predicted and actual values. A slope close to 1 not only indicates strong correlation but also reflects that the model does not consistently overestimate or underestimate the RUL. This illustration effectively conveys the distribution and the spread of prediction errors, making it easier to visually assess the overall prediction quality and bias.

The predicted RUL well compliments the actual lathe spindle life degradation pattern. The LSTM architecture is well efficient in digging up hidden patterns from time-series data. Further employing a bi-LSTM network can refine the learned degradation patterns yielding accurate estimations for prognostic regression analysis using time-series sequence data.

A comparative evaluation of three evolved deep learning models, LSTM, bi-LSTM, and a hybrid LSTM + bi-LSTM predictive was conducted using RMSE and A20-index on two separate test datasets, as summarized in Table 4. The LSTM model demonstrates the most balanced and

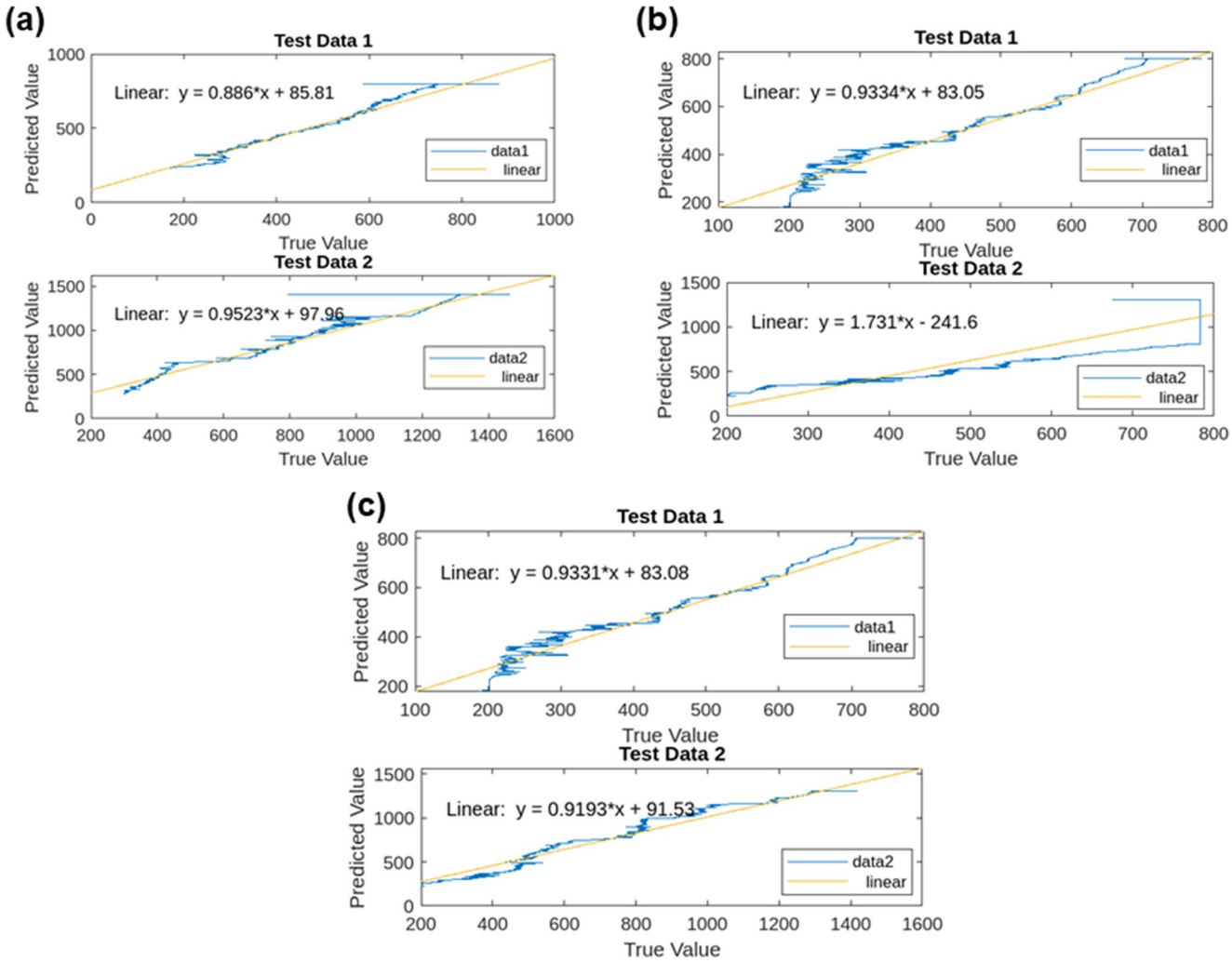

Fig. 8 Scatter Plots of predicted RUL versus actual RUL values (**a**) LSTM (**b**) bi-LSTM, and (**c**) LSTM + bi-LSTM network architecture

Table 4 Predictive model performance comparison metrics

Model	RMSE (in time-step)	A20-index Test Data 1	A20-index Test Data 2	Interpretation
LSTM	40.01	97.30%	92.57%	Best overall balance of accuracy and consistency
bi-LSTM	52.73	86.00%	48.68%	High error and inconsistent generalization
LSTM + bi-LSTM	31.65	86.00%	99.72%	Lowest error, but performance varies across datasets

consistent performance, achieving an RMSE of 40.01 time-steps, along with high A20-index values of 97.30% on Test Data 1 and 92.57% on Test Data 2. This indicates that a vast majority of its predictions fall within ±20% of the experimental values across both datasets. The bi-LSTM model shows the weakest performance, with the highest RMSE (52.73 time-steps) and a dramatic drop in A20-index from 86.00% to 48.68%, highlighting its lack of stability and poor generalization. The hybrid LSTM + bi-LSTM model achieves the lowest RMSE (31.65 time-steps) and excellent accuracy on Test Data 2 (A20 = 99.72%), but its A20-index on Test Data 1 (86.00%) is significantly lower than that of the LSTM model. While the hybrid model excels in minimizing average error and performs exceptionally on one dataset, its cross-dataset consistency is limited. Therefore, when considering both prediction accuracy and generalization across datasets, the LSTM model emerges as the most reliable and suitable choice for robust predictive tasks. Although, the bi-LSTM networks can enhance learning by capturing both forward and backward temporal dependencies, in the case of CNC lathe spindle degradation data, the

LSTM architecture outperforms bi-LSTM likely due to the strongly unidirectional nature of the mechanical degradation process. Spindle wear progresses over time in a forward direction, with past states influencing future behavior, but not vice versa. As a result, the backward context leveraged by bi-LSTM may introduce irrelevant or noisy dependencies, reducing generalization.

To address the possibility of overfitting during the RUL estimation process, a carefully designed feature processing and model regularization strategy was implemented. Initially, vibration signals obtained from the experimental run-to-failure setup were processed to extract a diverse set of statistical and signal-based features from the time, frequency, and time–frequency domains. To ensure that only the most informative features contributed to the model, NCA was employed as a supervised feature selection method. This helped reduce input dimensionality, remove redundancy, and preserve only those features that were most relevant to the spindle degradation patterns. By minimizing the complexity of the input space, the model was less prone to overfitting, especially given the limited dataset size. Furthermore, during model training, the deep learning hyperparameters dropout rate and regularization factor were incorporated into the LSTM/bi-LSTM architectures to further enhance generalization and robustness. The experimental data was collected under controlled lab conditions using an accelerated run-to-failure test rig. While this setup ensures repeatability, it is limited to fixed speeds and a single dominant failure mode. These constraints influence the model extended to more varied industrial conditions.

Lathe Spindle RUL Estimation and User Interface for Predictive Analytics

The RUL estimation required the lathe spindle real-time health status information and the evolved intelligent predictive model. The real-time lathe spindle health degradation data stored in the 'Lathe Spindle Health Status' channel is read using the 'Read API key' and the intelligent predictive model stored in a 'Drop Box' cloud storage is called using the 'DropBox Access Token', which the ''Drop Box API key'. The RUL life estimation algorithm first retrieves the intelligent predictive model and then the features representing the real-time health status of the lathe spindle.

The algorithm predicts the RUL of the lathe spindle by analyzing the current health status information and historical health degradation patterns. The current health status of the lathe spindle is revealed through the extracted vibration signature features standard deviation, peak-to-peak, RMS, RSSq, energy, normal entropy, and log entropy, which are made available in the cloud space. The historical health degradation patterns of the lathe spindle are represented through the evolved LSTM prognostic regression model. The degradation pattern of the features is analyzed against the evolved prognostic regression model to estimate the RUL of the lathe spindle unit. The algorithm automatically executes at regular intervals of time and also can be executed at any time in the interest of the user.

Although the LSTM and bi-LSTM models involve considerable complexity during training, only the final trained model is deployed on the edge device for real-time inference. This significantly reduces computational demands, as inference requires far less processing power compared to model training. As a result, the hybrid model can be effectively implemented on resource-constrained edge devices without notable performance issues. The ThingSpeak™ cloud platform is used primarily for data monitoring, visualization, and alerting, while the predictive computation is handled locally. This architecture balances model performance with deployment feasibility in industrial environments.

The determined RUL values are then used as a facet to trigger the failure warning system through email alerts. The system is designed to evaluate the RUL at a regular interval of time and if the obtained RUL value crosses a predefined safe threshold, the system automatically triggers the email alerts. The email alert warning includes a message on if the machine is running well or if the machine requires maintenance. The email also provides the available RUL of the spindle unit, which also provides the user to make a suitable decision regarding machinery maintenance and planning.

The outputs of the proposed IoT architecture are integrated into a web page user interface that allows the user to have live monitoring of lathe spindle health status from anywhere in the world. The live lathe spindle health degradation data sent to the ThingSpeak channel is reflected as such in the webpage interface on a real-time scale. Figure 9 shows screenshots on the developed webpage user interface.

Figure 9(a) displays the introduction page detailing the identity and purpose of the webpage. A live health status indicator dial gauge is also designed and integrated into the webpage for easy monitoring of lathe spindle health status. Figure 9 (b) displays the live health status indicator dial gauge that has three stages of dial gauge indicating the safe (Green), alert (Yellow), and faulty (Red) zones of lathe spindle operational lifetime. The page also displays the health degradation pattern of the current operating lathe spindle. The interface can be utilized as a real-time maintenance decision-making dashboard for the lathe machine tool system.

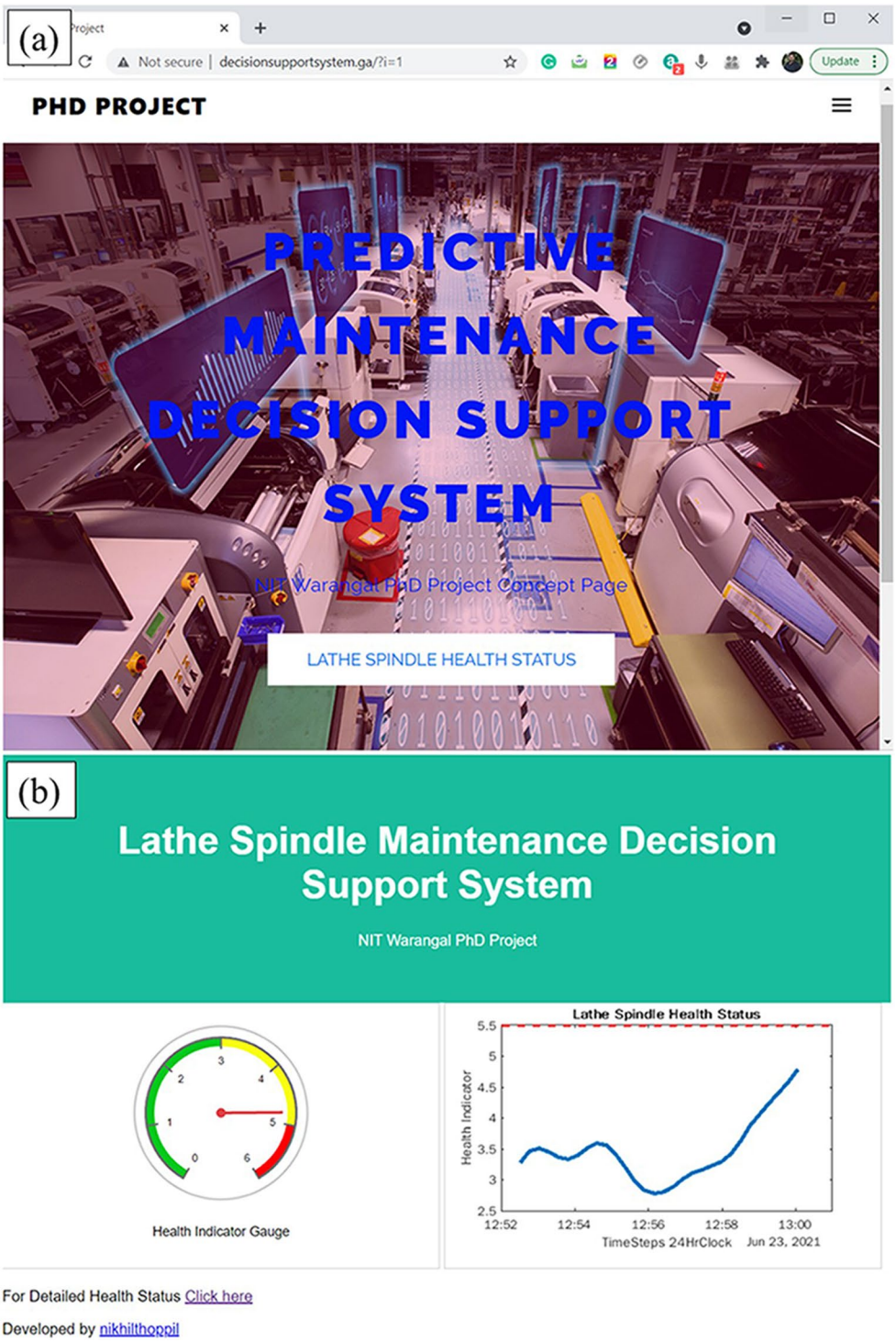

Fig. 9 Screenshot of the developed webpage interface (**a**) cover page, (**b**) Health Indicator dial gauge and degradation trend

Conclusions

In conclusion, this work presents an efficient and effective framework for RUL estimation, real-time machinery health status monitoring, and predictive maintenance decision-making of CNC lathe spindle unit. By integrating advanced digital technologies like IIoT, data-driven predictive analytics, and cloud computing, this framework supports proactive maintenance strategies through predictive analytics. The proposed system ensures that machine health is continuously monitored and maintenance decisions are automated, minimizing downtime and optimizing operational efficiency. The findings also highlight the feasibility of deploying deep learning models for real-time inference

on edge devices, supported by cloud-based visualization and alerting. The conclusion made from the study can be summarized as follows:

i. The ThingSpeak IoT web service platform and DropBox cloud storage services are utilized for developing the IoT-based real-time maintenance decision-making dashboard.
ii. The real-time lathe spindle monitoring vibration data is analyzed to extract signature features and then the selected features are simultaneously sent to a cloud space storage on a real-time basis.
iii. The features representing the current health status of the lathe spindle are analyzed against the trained predictive model to estimate the real-time RUL of the lathe spindle unit. This RUL value crossing a predefined threshold automatically triggers email warnings to alert the user regarding the maintenance of the lathe machine tool.
iv. The real-time lathe spindle health status information is also made available in a webpage user interface, which allows the user to access the machinery live health status information from anywhere in the world.

This work showcases how the integration of IIoT, cloud computing, and deep learning can enhance industrial maintenance strategies, providing a robust, scalable solution for real-time health monitoring and predictive maintenance of CNC machine tools. The adoption of such systems can significantly reduce maintenance costs and improve the operational longevity of critical manufacturing equipment.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42417-025-02144-6.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability Not Applicable.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

1. Soori M, Arezoo B, Dastres R (2023) Machine learning and artificial intelligence in CNC machine tools: a review. Sustainable Manufacturing and Service Economics 2:100009. https://doi.org/10.1016/j.smse.2023.100009
2. Yao X, Zhou J, Lin Y, Li Y, Yu H, Liu Y (2019) Smart manufacturing based on cyber-physical systems and beyond. J Intell Manuf 30(8):2805–2817. https://doi.org/10.1007/s10845-017-1384-5
3. Butler Q, Ziada Y, Stephenson D, Gadsden A (2022) S, Condition Monitoring of Machine Tool Feed Drives: A Review. ASME. J. Manuf. Sci. Eng. October 2022; 144(10): 100802 https://doi.org/10.1115/1.4054516
4. Cen J, Yang Z, Liu X et al (2022) A review of data-driven machinery fault diagnosis using machine learning algorithms. J Vib Eng Technol 10:2481–2507. https://doi.org/10.1007/s42417-022-00498-9
5. Suvarna M, Yap KS, Yang W, Li J, Ng YT, Wang X (2021) Cyber–physical production systems for data-driven, decentralized, and secure manufacturing—a perspective. Engineering 7(9):1212–1223. https://doi.org/10.1016/j.eng.2021.04.021
6. Dalzochio J, Kunst R, Pignaton E, Binotto A, Sanyal S, Favilla J, Barbosa J (2020) Machine learning and reasoning for predictive maintenance in industry 4.0: current status and challenges. Comput Ind 123:103298. https://doi.org/10.1016/j.compind.2020.103298
7. Zonta T, da Costa CA, da Rosa Righi R, de Lima MJ, da Trindade ES, Li GP (2020) Predictive maintenance in the industry 4.0: A systematic literature review. Comput Ind Eng 150:106889. https://doi.org/10.1016/j.cie.2020.106889
8. Murtaza AA, Saher A, Zafar MH, Moosavi SK, R, Aftab MF, Sanfilippo F (2024) Paradigm shift for predictive maintenance and condition monitoring from industry 4.0 to industry 5.0: A systematic review, challenges and case study. Results Eng 24:102935. https://doi.org/10.1016/j.rineng.2024.102935
9. Irgat E, Çinar E, Ünsal A et al (2023) An IoT-based monitoring system for induction motor faults utilizing deep learning models. J Vib Eng Technol 11:3579–3589. https://doi.org/10.1007/s42417-022-00769-5
10. Aboshosha A, Haggag A, George N et al (2023) IoT-based data-driven predictive maintenance relying on fuzzy system and artificial neural networks. Sci Rep 13:12186. https://doi.org/10.1038/s41598-023-38887-z
11. Patil A, Soni G, Prakash A (2024) Data-driven approaches for impending fault detection of industrial systems: a review. Int J Syst Assur Eng Manag 15:1326–1344. https://doi.org/10.1007/s13198-022-01841-9
12. Nunes P, Santos J, Rocha E (2023) Challenges in predictive maintenance – a review. CIRP J Manuf Sci Technol 40:53–67. https://doi.org/10.1016/j.cirpj.2022.11.004
13. Chataut R, Phoummalayvane A, Akl R (2023) Unleashing the power of IoT: a comprehensive review of IoT applications and future prospects in healthcare, agriculture, smart homes, smart cities, and industry 4.0. Sensors 23(16):7194. https://doi.org/10.3390/s23167194
14. Wu K, Xu J, Zheng M (2024) Industry 4.0: review and proposal for implementing a smart factory. Int J Adv Manuf Technol 133:1331–1347. https://doi.org/10.1007/s00170-024-13839-7
15. Asteris PG, Gavriilaki E, Touloumenidou T et al (2022) Genetic prediction of ICU hospitalization and mortality in COVID-19 patients using artificial neural networks. J Cell Mol Med 26:1445–1455. https://doi.org/10.1111/jcmm.17098
16. Mahmood W, Mohammed AS, Asteris PG, Kurda R, Armaghani DJ (2022) Modeling flexural and compressive strengths behaviour of cement-grouted sands modified with water reducer polymer. Appl Sci 12(3):1016. https://doi.org/10.3390/app12031016
17. Peter O, Pradhan A, Mbohwa C (2023) Industrial internet of things (IIoT): opportunities, challenges, and requirements in manufacturing businesses in emerging economies. Procedia Comput Sci 217:856–865. https://doi.org/10.1016/j.procs.2022.12.282
18. Khalil RA, Saeed N, Masood M, Fard YM, Alouini M-S, Al-Naffouri TY (2021) Deep learning in the industrial internet of things:

potentials, challenges, and emerging applications. IEEE Internet Things J 8(14):11016–11040. https://doi.org/10.1109/JIOT.2021.3051414
19. Resende C, Folgado D, Oliveira J, Franco B, Moreira W, Oliveira-Jr A, Cavaleiro A, Carvalho R (2021) TIP4.0: industrial internet of things platform for predictive maintenance. Sensors (Basel) 21(14):4676. https://doi.org/10.3390/s21144676
20. Daji D, Ghule K, Gagdani S, Butala A, Talele P, Kamat H (2020) Cloud-based asset monitoring and predictive maintenance in an industrial IoT system. In Proceedings of the 2020 International Conference for Emerging Technology (INCET) (pp. 1–5). IEEE. https://doi.org/10.1109/INCET49848.2020.9154148
21. Mohapatra AG, Mohanty A, Tripathy PK (2024) IoT-enabled predictive maintenance and analytic hierarchy process-based prioritization of real-time parameters in a diesel generator: an industry 4.0 case study. SN Comput Sci 5:145. https://doi.org/10.1007/s42979-023-02508-3
22. Hadi RH, Hady HN, Hasan AM, Al-Jodah A, Humaidi AJ (2023) Improved fault classification for predictive maintenance in industrial IoT based on automl: A case study of Ball-Bearing faults. Processes 11(5):1507. https://doi.org/10.3390/pr11051507
23. Hafeez T, Xu L, McArdle G (2021) Edge intelligence for data handling and predictive maintenance in IIoT. IEEE Access 9:49355–49371. https://doi.org/10.1109/ACCESS.2021.3069137
24. Cakir M, Guvenc MA, Mistikoglu S (2021) The experimental application of popular machine learning algorithms on predictive maintenance and the design of IIoT based condition monitoring system. Comput Ind Eng 151:106948. https://doi.org/10.1016/j.cie.2020.106948
25. Li W, Law KL, E (2024) Deep learning models for time series forecasting: A review. IEEE Access 12:92306–92327. https://doi.org/10.1109/ACCESS.2024.3422528
26. Ali Y, Hussain F, Haque MM (2024) Advances, challenges, and future research needs in machine learning-based crash prediction models: a systematic review. Accid Anal Prev 194:107378. https://doi.org/10.1016/j.aap.2023.107378
27. Emmert-streib F, Yang Z, Feng H, Tripathi S, Emmert-streib F (2020) An introductory review of deep learning for prediction models with big data. Front Artif Intell 3:1–23. https://doi.org/10.3389/frai.2020.00004
28. Thoppil NM, Vasu V, Rao CSP (2021) Deep learning algorithms for machinery health prognostics using time-series data: a review. J Vib Eng Technol 9:1123–1145. https://doi.org/10.1007/s42417-021-00286-x
29. Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735
30. Houdt GV, Mosquera C, Nápoles G (2020) A review on the long short-term memory model. Artif Intell Rev 53:5929–5955. https://doi.org/10.1007/s10462-020-09838-1
31. Kumar A, Parey A, Kankar PK (2024) A new hybrid LSTM-GRU model for fault diagnosis of polymer gears using vibration signals. J Vib Eng Technol 12(2):2729–2741. https://doi.org/10.1007/s42417-023-01010-7
32. Foumani NM, Miller L, Tan CW, Webb GI, Forestier G, Salehi M (2024) Deep learning for time series classification and extrinsic regression: a current survey. ACM Comput Surv 56:45 9. https://doi.org/10.1145/3649448
33. Zou L, Ma C, Hu J, Yu Z, Zhuang K (2024) Enhanced predictive modeling of rotating machinery remaining useful life by using separable convolution backbone networks. Appl Soft Comput 156:111493. https://doi.org/10.1016/j.asoc.2024.111493
34. Qi J, Zhu R, Liu C, Mauricio A, Gryllias K (2024) Anomaly detection and multi-step estimation based remaining useful life prediction for rolling element bearings. Mech Syst Signal Process 206:110910. https://doi.org/10.1016/j.ymssp.2023.110910
35. Pei X, Gao L, Li X (2024) Remaining useful life prediction of machinery based on performance evaluation and online cross-domain health indicator under unknown working conditions. J Manuf Syst 75:213–227. https://doi.org/10.1016/j.jmsy.2023.12.004
36. He M, Zhou Y, Li Y, Wu G, Tang G (2020) Long short-term memory network with multi-resolution singular value decomposition for prediction of bearing performance degradation. Measurement J Int Meas Confederation 156:107582. https://doi.org/10.1016/j.measurement.2020.107582
37. Tseng SH, Tran KD (2024) Predicting maintenance through an attention long short-term memory projected model. J Intell Manuf 35:807–824. https://doi.org/10.1007/s10845-023-02077-5
38. Fawaz HI, Forestier G, Weber J, Idoumghar L (2019) Deep learning for time series classification: a review. Data Min Knowl Discov 33(4):917–963. https://doi.org/10.1007/s10618-019-00619-1
39. Thoppil NM, Vasu V, Rao CSP (2020) On the criticality analysis of computer numerical control lathe subsystems for predictive maintenance. Arab J Sci Eng 45:5259–5271. https://doi.org/10.1007/s13369-020-04397-7
40. Hu L, Pueh LH, Wang D, Wang Z (2024) Fatigue failure of high precision spindle bearing under extreme service conditions. Eng Fail Anal 158:107951. https://doi.org/10.1016/j.engfailanal.2023.107951
41. Jaber AA (2024) Diagnosis of bearing faults using temporal vibration signals: a comparative study of machine learning models with feature selection techniques. J Fail Anal Prev 24:752–768. https://doi.org/10.1007/s11668-024-01883-0
42. Hashemi SM, Sambandamurthy H, Ghaemi H (2015) Vibration analysis of machine tool spindle systems: A calibrated finite element model. IMECE. V04BT04A077 ASME, Montreal, Quebec, Canada https://doi.org/10.1115/IMECE2014-39547
43. Patil AR, Buchaiah S, Shakya P (2024) Combined VMD-Morlet wavelet filter based signal de-noising approach and its applications in bearing fault diagnosis. J Vib Eng Technol 12:7929–7953. https://doi.org/10.1007/s42417-024-01338-8
44. Abbasimehr H, Shabani M, Yousefi M (2020) An optimized model using LSTM network for demand forecasting. Comput Ind Eng 143:106435. https://doi.org/10.1016/j.cie.2020.106435
45. Thoppil NM, Vasu V, Rao CS, P (2021) Bayesian Optimization LSTM and bi-LSTM Network with Self-Optimized Structure and Hyperparameters for Remaining Useful Life Estimation of Lathe Spindle Unit. ASME. J. Comput. Inf. Sci. Eng. April 2022; 22(2): 021012. https://doi.org/10.1115/1.4052838
46. Zhang C, Mousavi AA, Masri SF, Gholipour G, Yan K, Li X (2022) Vibration feature extraction using signal processing techniques for structural health monitoring: a review. Mech Syst Signal Process 177:109175. https://doi.org/10.1016/j.ymssp.2022.109175
47. Pandiyan M, Babu TN (2024) Systematic review on fault diagnosis on rolling-element bearing. J Vib Eng Technol 12:8249–8283. https://doi.org/10.1007/s42417-024-01358-4
48. Asteris PG, Karoglou M, Skentou AD, Vasconcelos G, He M, Bakolas A, Zhou J, Armaghani DJ (2024) Predicting uniaxial

compressive strength of rocks using ANN models: incorporating porosity, compressional wave velocity, and Schmidt hammer data, ultrasonics. 141:107347. https://doi.org/10.1016/j.ultras.2023.107347

49. Asteris PG, Tsavdaridis KD, Lemonis ME et al (2024) AI-powered GUI for prediction of axial compression capacity in concrete-filled steel tube columns. Neural Comput Appl 36:22429–22459. https://doi.org/10.1007/s00521-024-10405-w
50. Kamalakkannan S, Kulatunga AK (2022) IIOT applications for sustainable manufacturing. In: Hussain CM, Di Sia P (eds) Handbook of smart Materials, Technologies, and devices. Springer, Cham. https://doi.org/10.1007/978-3-030-84205-5_31
51. Ganga SMS, Shiva PMS, Rao BB, Chetti S, Naikodi C (2024) Integration of NIST-Aligned Secure IoT Data Management Framework Tinkercad, Flask, and ThingSpeak, 2024 First International Conference on Innovations in Communications, Electrical and Computer Engineering (ICICEC), Davangere, India, pp. 1–8. https://doi.org/10.1109/ICICEC62498.2024.10808927

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.